

SAGE UGRAS

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EDUCATION

Worcester Polytechnic Institute

BS Mechanical and Materials Engineering - GPA: 3.92/4.0 (High Honors)
WPI Presidential Scholar | Dean's List (All Semesters)

Aug. 2021 to May 2025

Worcester, MA

EXPERIENCE

Wisk Aero

Mechanical Engineer - Systems Test

Mountain View, CA

July 2025 to Present

Hired full-time post-internship and earned a promotion within the first year. Serving as the sole mechanical engineer on the battery test team, owning end-to-end fixture design and test operations from concept through execution.

- Designed custom test fixtures and Poka-Yoke tools for PCBAs and battery modules, eliminating operator assembly errors and reducing test setup cycle times.
- Enabled flight operations by executing HV battery vibration testing on the critical path.
- Develop and commission an instrumented coolant distribution assembly for CFD thermal model improvement.
- Serve as NX CAD superuser, establishing modeling standards and mentoring team members on assembly design and troubleshooting.
- Collaborate with thermal and systems engineering teams to execute test procedures and validate battery performance.
- Monitor real-time telemetry from the Power and Propulsion seat during live flight test operations to ensure hazard mitigation and safety compliance.
- Mentor interns on high-voltage safety protocols, mechanical design, and test bench operation.

Systems Test Intern

June 2024 to Aug. 2024

- Developed Gen 6 battery module acceptance testing, authoring initial ATP documentation and physically prototyping, assembling, and validating early-stage hardware.
- Designed and commissioned a mobile module charging station integrated with automated contactors, power supplies, and hardwired E-stop circuits, enabling safe off-site and production battery testing.
- Built custom high-voltage cables and breakout boxes to connect battery modules, PCBAs, and test equipment across bench setups.
- Instrumented high-density thermistor arrays for thermal runaway testing, capturing key thermal propagation metrics for safety qualification.

Our Next Energy

Product Engineering Intern

Novi, MI

May 2023 to Aug. 2023

- Conceptualized, prototyped, and tested a vacuum-assisted busbar alignment fixture to resolve a production line bottleneck, presenting models directly to the CEO on the shop floor.
- Designed the structural pack enclosure backstop for the company's flagship grid energy storage R&D prototype.
- Engineered custom laser cutter fixturing for precise insulation removal on battery cells, reducing processing time by 80%.

PROJECTS

Major Qualifying Project (Formula SAE Electric Team)

Aug. 2024 to June 2025

- Led vehicle integration and CAD reviews, co-redesigning accumulator modules for enhanced structural integrity and manufacturability.
- Designed an ergonomic steering wheel with integrated mechanical paddle shifters, PCBs, and ingress protection.
- Created manufacturing process for composite aero fairings using custom resin infusion and a custom hot-wire cutter.

Independent Study - Accumulator Design (Formula SAE Electric Team)

Oct. 2023 to Mar. 2024

- Designed and manufactured mechanical retention and busbar structural components for high-voltage (345.6Vnom) vehicle battery pack.

ACTIVITIES

WPI Bionics Club - President

Aug. 2022 to May 2025

- Directed operations and mechanical development for open-source prosthetic limbs.

WPI Collablab -Vice President

Aug. 2022 to May 2025

- Managed makerspace operations and trained 40+ members on safety protocols, 3D printing, and machining.

SKILLS

Design: NX, Solidworks, Onshape, Altium, Fusion 360, Capitol Harness Designer

Manufacturing: 3D Printing, Laser Cutting, Manual Machining, Waterjet Cutting, CNC Machining

Software: Arduino, Python, Data Acquisition and Logging Programs, Crystal Instruments EDM

CERTIFICATIONS

SAE Safe Handling of High Voltage Battery Systems | Designing Parts in NX for Experienced Users 2306 | RTCA DO-160G | OSHA-10